

Appendix A - Requirements Engineering Risks & Risk Management Plan

for

docWise

Prepared by Rukiye Sedef Keskin

Supervisor: Assoc. Prof. EBRU GÖKALP AYDIN
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Docwise Organization

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1. Introduction

This document has been prepared to identify and manage potential risks that may arise during the requirements engineering process for the AI-Assisted Diagnosis System. The **risk management plan** is critically important for enhancing the **reliability, accuracy, and compliance** of the system.

2. Risk Identification

Below risks are categorized as **technical, security, user acceptance, operational, and ethical**.

2.1. Technical Risks

Risk	Description	Risk Level (H/M/L)
AI making incorrect diagnoses	The model producing biased results due to training with incorrect data	H
Incomplete or inaccurate datasets	Lack of diversity in training data	M
Insufficient system performance	Increased response time when handling large patient datasets	M
Delays in cloud-based data access	Potential delays when retrieving data from remote storage systems	L
Hardware incompatibility	AI model not integrating with legacy hospital systems	M
Instability in model updates	New updates leading to unexpected errors	H

2.2. Security and Regulation Risks

Risk	Description	Risk Level (H/M/L)
Violation of patient data privacy	Non-compliance with data security standards such as GDPR and HIPAA	H

Risk	Description	Risk Level (H/M/L)
Unauthorized access to the system	Unauthorized access to patient data	H
Data leaks	Large datasets being compromised by third parties	H
Malicious software attacks	Hackers attacking the AI system to cause harm	H

2.3. User Acceptance Risks

Risk	Description	Risk Level (H/M/L)
Doctors resisting the AI system	Lack of trust in AI diagnoses	H
Patients not trusting AI diagnoses	Concerns that AI is making decisions instead of humans	M

2.4. Operational Risks

Risk	Description	Risk Level (H/M/L)
Service disruption during system updates	Patients or doctors unable to receive diagnostic support	M
Healthcare providers struggling to use the AI system	Users having difficulty understanding the system	M

2.5. Ethical Risks

Risk	Description	Risk Level (H/M/L)
AI making biased diagnoses against certain patient groups (biased training data)	The training data used in the AI model is skewed or biased	H
AI diagnostic results causing harm to patients	Patients' health conditions worsening due to incorrect diagnoses	H

3. Risk Response Strategies

Risk	Strategy	Owner (Responsible Person)	Action Plan
AI making incorrect diagnoses	Mitigate	SAr	Regularly update the AI model, introduce doctor approval for diagnoses, and implement a self-improvement mechanism.
Incomplete or inaccurate datasets	Mitigate	SAn	Improve training data quality by incorporating diverse data sources.
Insufficient system performance	Mitigate	SAr	Use high-performance servers and continuously test system optimization.
Delays in cloud-based data access	Accept	CM	Accept up to a certain level, but implement caching and fast access systems for critical scenarios.
Hardware incompatibility	Mitigate	SAr	Develop a modular architecture for AI models to work with different systems.
Instability in model updates	Mitigate	SAr	Use A/B testing for updates and test them with a small user group before deployment.
Violation of patient data privacy	Avoid	CM	Apply strong encryption protocols and ensure full compliance with GDPR and HIPAA regulations.
Unauthorized access to the system	Transfer	CM	Security management should be provided by a cloud provider and IT team, enforcing two-factor authentication (TFA) .
Data leaks	Mitigate	CM	Implement regular security scans to detect unauthorized access and restrict access permissions.
Malicious software attacks	Avoid	CM	Apply strong firewalls and anti-malware security policies .
Doctors resisting the AI system	Mitigate	SAn	Conduct training programs explaining the benefits of the AI system.
Patients not trusting AI diagnoses	Mitigate	SAn	Educate patients on how AI works with doctors and that it does not make independent decisions.
Service disruption during system updates	Mitigate	CM	Perform updates during off-peak hours and when there are fewer users.

Risk	Strategy	Owner (Responsible Person)	Action Plan
Healthcare providers struggling to use the AI system	Mitigate	SAn	Design a user-friendly interface and provide training to speed up adaptation.
Insurance companies rejecting AI diagnoses	Mitigate	PM	Work with insurance companies to prove the reliability of AI diagnoses.
Regulatory changes	Accept	PM	Obtain legal consultation to quickly adapt to regulatory changes.
AI making biased diagnoses against specific patient groups	Mitigate	SAr	Ensure the AI model passes ethical audits and train it with diverse datasets .
AI diagnostic results causing harm to patients	Mitigate	ST	Require doctor approval for diagnoses and continuously update the system to prevent misdiagnoses.

4. Risk Monitoring and Mitigation

The following methods will be implemented for the **continuous monitoring and mitigation of risks**:

4.1. Risk Monitoring and Early Warning System

- **AI diagnosis accuracy analysis** will be continuously monitored.
- A **user feedback mechanism** will collect opinions from patients and doctors about AI diagnoses.
- **Potential system errors** will be recorded through an automatic logging system.

4.2. Feedback Mechanism and User Participation

- **AI diagnosis accuracy analysis** will be continuously monitored.
- A **user feedback mechanism** will collect opinions from patients and doctors about AI diagnoses.
- **Potential system errors** will be recorded through an automatic logging system.

4.3. Security Protocol Updates

- **Two-Factor Authentication (TFA)** will be used to prevent unauthorized access.
- **Patient data will be protected using the latest encryption methods.**
- **Access logs** will be monitored in real-time to detect unauthorized entries.

4.4. Regulatory Compliance Audits

- The system will ensure compliance with **health data security regulations such as KVKK and HIPAA.**
- Regular meetings will be conducted with **hospitals, insurance companies, and government institutions** to ensure legal compliance.

4.5. Model Updates and Performance Testing

- Updates that **reduce incorrect diagnoses** will be applied regularly to the AI model.
- **New updates will be released only after successful A/B testing.**
- AI models tested with real patient data will **only operate under doctor supervision.**